Cohomological Rigidity of Moment Angle Complexes

Labix Liu

Queen Mary University of London

Introduction and Goals

Given a simplicial complex $\mathcal{K}$, define the moment angle complex associated to $\mathcal{K}$ to be

$$\mathcal{Z}_{\mathcal{K}} = \bigcup_{\sigma \in \mathcal{K}} (D^2, S^1)^{\sigma}$$

where

$$(D^2, S^1)^{\sigma} = \prod_{i=1}^{n} Y_i \quad \text{where} \quad Y_i = \begin{cases} D^2 & \text{if } i \in \sigma \\ S^1 & \text{otherwise} \end{cases}$$

The cohomological rigidity conjecture states that for two simplicial complexes $\mathcal{K}_1$ and $\mathcal{K}_2$, we have

$$H^*(\mathcal{Z}_{\mathcal{K}_1}) \cong H^*(\mathcal{Z}_{\mathcal{K}_2})$$

if and only if $\mathcal{Z}_{\mathcal{K}_1}$ and $\mathcal{Z}_{\mathcal{K}_2}$ are diffeomorphic / homeomorphic / homotopy equivalent.

The main focus of our work is to determine the homotopy type of moment angle complexes arising from simplicial 4-polytopes and 8-vertices.

The Foundations

To tackle the cohomological rigidity conjecture we need to first understand the cohomology. Hochster's formula in [BP14] decomposes the cohomology into the cohomology of the simplicial complex itself. Namely, there is an isomorphism of rings

$$H^*(\mathcal{Z}_{\mathcal{K}}) \cong \bigoplus_{J \subseteq \{1,\dots,n\}} H^*(\mathcal{K}_J)$$

where

- $\mathcal{K}_J$ is the full subcomplex of $\mathcal{K}$ with the vertices in J.
- The additive isomorphism is given by
$$H^n(\mathcal{Z}_{\mathcal{K}}) \cong \bigoplus_{J \subseteq \{1,\dots,n\}} H^{n-|J|-1}(\mathcal{K}_J)$$
- The ring structure is given the map $H^i(\mathcal{K}_I) \otimes H^j(\mathcal{K}_J) \to H^{i+j+1}(\mathcal{K}_{I \cup H_J})$ which sends products of generators to generators if $I \cap J = \emptyset$ and 0 otherwise.

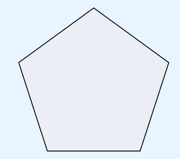
Now we need to understand the space itself. We list two usful results here.

- [McG79] states that for a stacked polytope P, there is a diffeomorphism
$$\mathcal{Z}_{\partial P} \cong \#_{k=3}^{m-n+1} (S^k \times S^{m+n-k})^{\#(k-2)\binom{m-n}{k-1}}$$
where $n = \dim(P)$ and $m = |V(P)|$.
- [ST26] states that for a simplicial complex $\mathcal{K}$ that is a neighbourly triangulation of S^{2n+1}, there is a homotopy equivalence
$$\mathcal{Z}_{\mathcal{K}} \simeq \#_{i=1}^{j} \left(S^{t_i} \times S^{s_i}\right)$$
for some $j \in \mathbb{N}^+$ and some $t_i, s_i \in \mathbb{N}^+$.

Notice that in both cases the diffeomorphism type and the homotopy type are both connected sums of products of two spheres.

An Example

Let us investigate the main techniques used in the proof of [ST26] using

$\mathcal{K} =$ 

1. Using Hochster's formula and [Hat02] 4C.1, we deduce that the 6-skeleton of the 7-dimensional moment angle complex is given by
$$\overline{\mathcal{Z}_{\mathcal{K}}} = \bigvee_{i=1}^{5} (S^3 \vee S^4)$$
2. The attaching map of the top cell is given by $S^6 \to \mathcal{Z}_{\mathcal{K}}$. The Hilton-Milnor theorem tells us
$$\pi_6(\overline{\mathcal{Z}_{\mathcal{K}}}) \cong \pi_6(S^3)^5 \times \pi_6(S^4) \times \pi_6(S^5)^{10} \times \pi_6(S^6)^{25}$$
3. In the theorem, our attaching map in the component $\pi_6(S^6)^{10}$ is of the form $S^6 \to S^6 \overset{\text{Wh}}{\to} S^3 \vee S^4 \hookrightarrow \overline{\mathcal{Z}_{\mathcal{K}}}$. Follow a similar argument in [Hat02] 4C.2 to read the map as cup products in the cohomology ring. Hochster's formula implies that the coefficients of this map is ± 1 in 5 cases and 0 in the others. This is precisely the attaching map for $(S^3 \times S^4)^{\#5}$.
4. In the theorem, our attaching map in the component $\pi_6(S^5)^{25}$ is of the form $S^6 \to S^5 \overset{\text{Wh}}{\to} S^3 \vee S^3 \hookrightarrow \overline{\mathcal{Z}_{\mathcal{K}}}$. Let $J \subseteq \{1, \dots, 5\}$ be a subset of four elements. Then we have
$$\mathcal{Z}_{\mathcal{K}_J} = S^3 \vee S^3 \vee S^3 \vee S^4 \vee S^4$$
There are five choices of J that contains all possible permutation of the choices of S^3's that make up the Whitehead product. Then using the retractions $\mathcal{Z}_{\mathcal{K}_J} \to \mathcal{Z}_{\mathcal{K}} \to \mathcal{Z}_{\mathcal{K}_J}$ and the null-homotopy $S^6 \to S^5 \to S^3 \vee S^3 \hookrightarrow \overline{\mathcal{Z}_{\mathcal{K}}} \to \mathcal{Z}_{\mathcal{K}}$ gives us the fact that $S^6 \to S^5$ is null homotopic. The remaining $\pi_6(S^3)$ and $\pi_6(S^4)$ follow a similar argument by choosing the appropriate subcomplexes.

The flow of the proof goes as follows:

$$\mathcal{K} \to \begin{array}{c}\text{Cohomology}\\ \text{of } \mathcal{Z}_{\mathcal{K}}\end{array} \to \begin{array}{c}\text{The top}-1\\ \text{skeleton of } \overline{\mathcal{Z}_{\mathcal{K}}}\end{array} \to \begin{array}{c}\text{Analyze the top attaching map}\\ \text{with Hilton-Milnor theorem}\end{array}$$

The Case of Simplicial 4-Polytopes with 8 Vertices

Simplicial 4-polytopes with 8 vertices are classified in [GS67] and are denoted by P_i^8 for $1 \leq i \leq 37$. The paper [OP24] says that for a simplicial complex $\mathcal{K}$ that is a triangulation of S^3, we have

$$H^*(\mathcal{Z}_{\mathcal{K}}) \cong H^*\left(\#_{i=1}^{j} \left(\prod_{k=1}^{t_i} S^{d_k}\right)\right)$$

if and only if exactly one of the following are true.

- $\mathcal{K} = S^0 * S^0 * S^0 * S^0$.
- The underlying graph of $\mathcal{K}$ is chordal.
- The underlying graph of $\mathcal{K}$ has exactly two missing edges with distinct vertices (forming a square).

Known Results

We list some of the known results here.

- P_1^8, P_2^8, P_3^8 are stacked polytopes, so it follows from the previous sections that their diffeomorphism type is identified:
$$\mathcal{Z}_{\partial P_1^8} \cong \mathcal{Z}_{\partial P_2^8} \cong \mathcal{Z}_{\partial P_3^8} \cong (S^3 \times S^9)^{\#6} \# (S^4 \times S^8)^{\#8} \# (S^5 \times S^7)^{\#3}$$
- $P_{35}^8, P_{36}^8, P_{37}^8$ are neighbourly polytopes, so it follows from the previous sections that their homotopy type is identified:
$$\mathcal{Z}_{\partial P_{35}^8} \simeq \mathcal{Z}_{\partial P_{36}^8} \simeq \mathcal{Z}_{\partial P_{37}^8} \simeq (S^5 \times S^7)^{\#16} \# (S^6 \times S^6)^{\#15}$$
- P_{34}^8 is a cross polytope, given by $P_{34}^8 = S^0 * S^0 * S^0 * S^0$. [FCMW14] shows that we have a homeomorphism
$$\mathcal{Z}_{\partial P_{34}^8} \cong S^3 \times S^3 \times S^3 \times S^3$$
- The cohomology for $\mathcal{Z}_{\partial P_{28}^8}$ is computed by [FCMW14] and [Iri16] determined its diffeomorphism type to be
$$\mathcal{Z}_{\partial P_{28}^8} \cong (S^3 \times S^3 \times S^6) \# (S^5 \times S^7)^{\#8} \# (S^6 \times S^6)^{\#8}$$

The Edge Cases

A quick check shows that other than $i = 7, 16, 17, 22, 26, 29$, the cohomology of $\mathcal{Z}_{\partial P_i^8}$ has the cohomology of a connected sum of a product of spheres. For the case $i = 7$, we computed the cohomology to be

$$H^*(\mathcal{Z}_{\partial P_7^8}) = H^*((S^3 \times S^9)^{\#3} \# (S^4 \times S^8)^{\#3} \# (S^5 \times S^7) \# X)$$

where

$$H^*(X) = \frac{\mathbb{Z}\langle a_1, a_2, b_1, b_2, c, d_1, d_2, e, f_1, f_2, g_1, g_2, h \rangle}{I}$$

and I is the ideal generated by the following relations:

- $a_1 a_2 = d_2$, $a_i b_i = e$, $a_i c = f_i$, $a_1 d_1 = g_2, a_2 d_2 = g_1$, $b_i c = g_i$.
- Poincare relations $a_i g_i = b_i f_i = ce = d_1 d_2 = h$.
- $z^2 = 0$ for all generators z.

and $\deg(a_i) = 3$, $\deg(b_i) = 4$, $\deg(c) = 5$, $\deg(d_i) = 6$, $\deg(e) = 7$, $\deg(f_i) = 8$, $\deg(h) = 12$ (13 generators and 28 relations).

References

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